

kAixu CodeStudio Pro - Technical Due Diligence & Valuation

Build 425ecf6c96d7 | Version 5.1.0 | 2026-02-22T10:40:41Z

This report analyzes the shipped client-only PWA build extracted from **kaixu-codestudio-f500-fullpatch.zip**. The focus is quantitative: codebase scale, security controls, enterprise gaps, and valuation ranges.

Scale

Total package size: **3.29 MB** across **22** files. Text/code is **111.5 KB** and **3,235** LOC; assets are **3.18 MB**. The largest single asset is **assets/logo.png** at **2.57 MB** (78.2% of total bytes).

Runtime dependencies

The build can run fully offline for editing, viewing, and local logs. AI calls require connectivity to exactly **one** external domain: Kaixu Gateway. The service worker pins **16** core assets for offline operation.

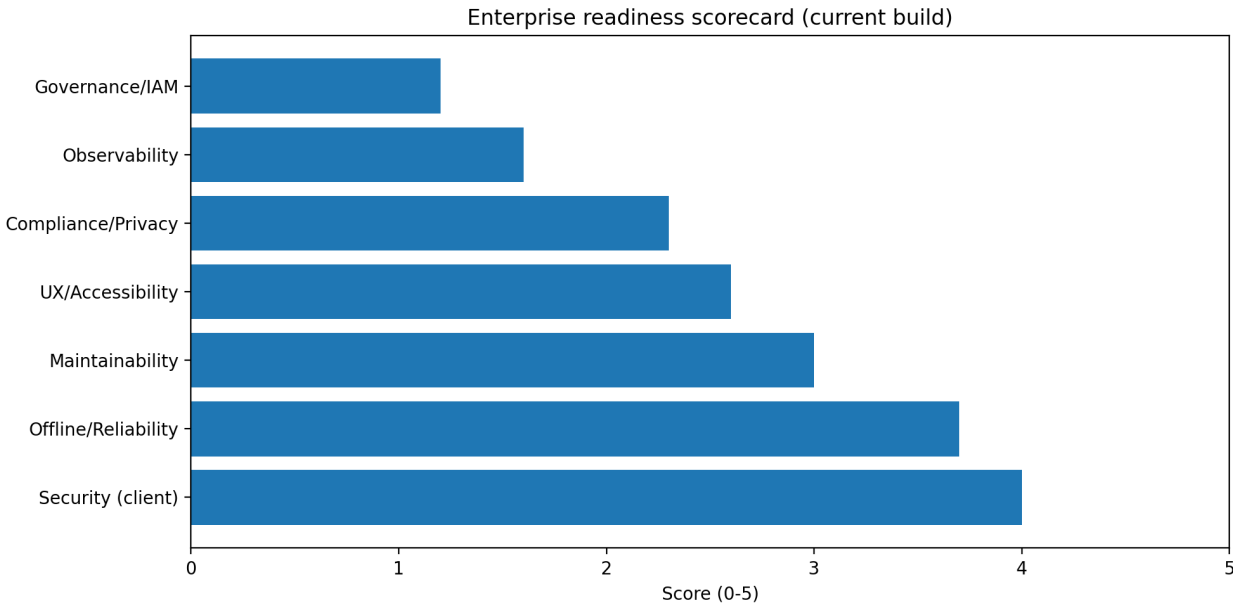
Security controls present

Strict CSP for the main app (no inline scripts and no unsafe-eval), AES-256-GCM encrypted vault with PBKDF2 derivation, optional workspace encryption at rest, and a sandbox runner whose CSP blocks network access.

Procurement reality

For a typical Fortune-500 rollout, the missing pieces are not more client-side CSP; they are centralized identity (SSO), RBAC, and server-side audit logging plus compliance evidence (e.g., SOC2 controls).

Enterprise readiness scorecard (0-5)



Score reflects enterprise procurement: strong local security/offline behavior; weak centralized governance/observability.

1) Build inventory and quantitative metrics

File count: **22**

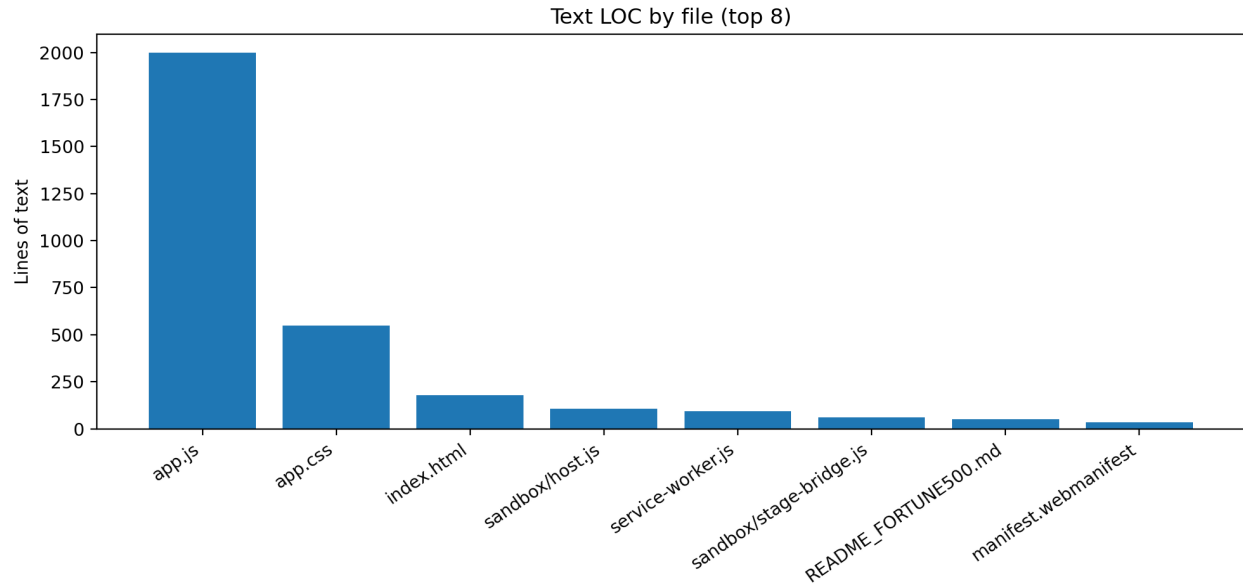
Total bytes: **3,291,167** (3.29 MB)

Total LOC (text files): **3,235**

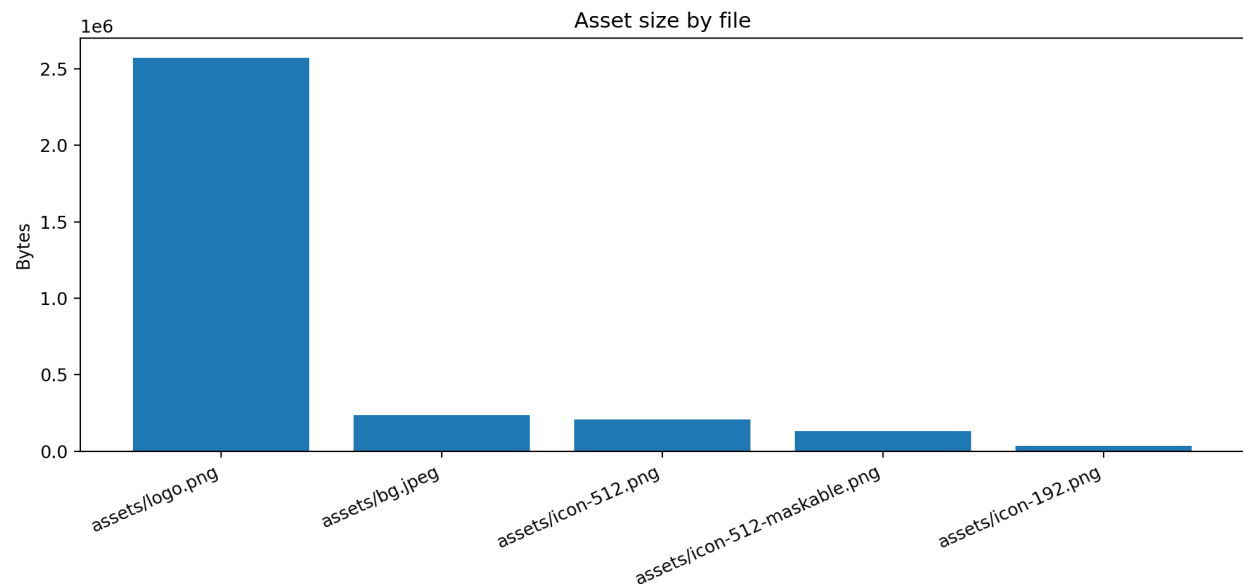
Gzipped critical code payload estimate: **27.9 KB** for HTML+CSS+JS (excludes images)

AI network calls: **1 POST** per request to <https://kaixugateway13.netlify.app/.netlify/functions/gateway-chat>

Text LOC distribution



Asset size distribution (performance risk)



assets/logo.png dominates package weight (78.2% of total bytes). This is the single biggest measurable performance issue.

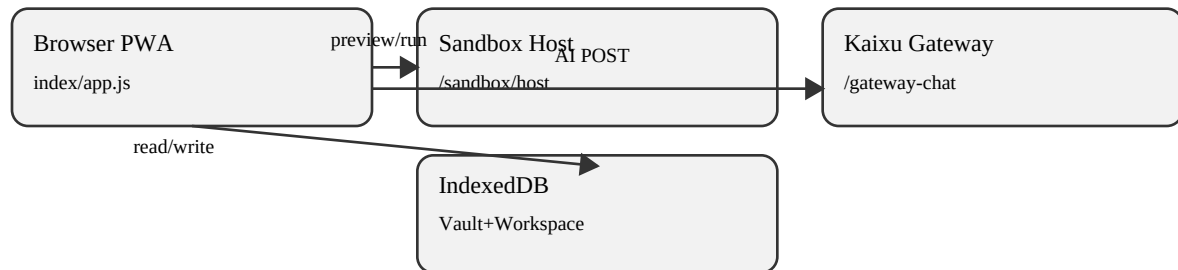
File inventory (top 16 by size)

File	Type	Bytes	LOC	Non-empty lines
assets/logo.png	.png	2,573,500		
assets/bg.jpeg	.jpeg	235,148		
assets/icon-512.png	.png	209,083		
assets/icon-512-maskable.png	.png	129,030		
app.js	.js	67,631	1998	1778
assets/icon-192.png	.png	32,902		
app.css	.css	18,672	549	519
index.html	.html	8,402	180	164
sandbox/host.js	.js	3,651	107	89
service-worker.js	.js	2,406	92	80
sandbox/stage-bridge.js	.js	1,906	62	55
README_FORTUNE500.md	.md	1,891	50	42
SECURITY.md	.md	1,430	32	25
_headers	_headers	1,394	20	18
offline.html	.html	1,070	35	33
THREAT_MODEL.md	.md	886	25	20

Total files in build: 22.

2) Architecture (as-built)

Static single-page PWA with IndexedDB persistence, cryptographic vault, sandboxed runner, and Kaixu Gateway for AI.



Architecture notes

Main app: index.html + app.js + app.css. Local storage: IndexedDB stores workspace, settings, audit log, and encrypted vault blob. Sandbox: /sandbox/host.html runs code in a sandboxed stage iframe; user JS is loaded via blob script URL (no eval). AI: app.js sends one POST request to the Kaixu Gateway endpoint; Authorization header uses the decrypted vault key in memory only.

Offline: service-worker pins 16 core assets and serves cached app shell during navigation failures. Workspace and logs remain usable offline; AI is disabled offline.

3) Security review (client-side)

CSP (main):

```
Content-Security-Policy: default-src 'self'; base-uri 'self'; object-src 'none';
frame-ancestors 'none'; form-action 'self'; img-src 'self' data: blob:; font-src
'self' data:; style-src 'self'; script-src 'self'; connect-src 'self'
https://kaixugateway13.netlify.app; frame-src 'self'; manifest-src 'self';
worker-src 'self'; media-src 'none'
```

CSP (sandbox):

```
Content-Security-Policy: default-src 'self'; base-uri 'none'; object-src 'none';
frame-ancestors 'self'; form-action 'none'; img-src 'self' data: blob:; font-src
'self' data:; style-src 'self' 'unsafe-inline'; script-src 'self' blob:;
connect-src 'none'; frame-src 'self' data: blob:; worker-src 'none'; media-src
'none'
```

Key controls present: gateway-only routing; no eval/new Function in executable app code (static scan); clickjacking blocked; secrets encrypted at rest (AES-256-GCM) with PBKDF2; vault autolock; sandbox blocks network (connect-src none).

Risk register (top issues)

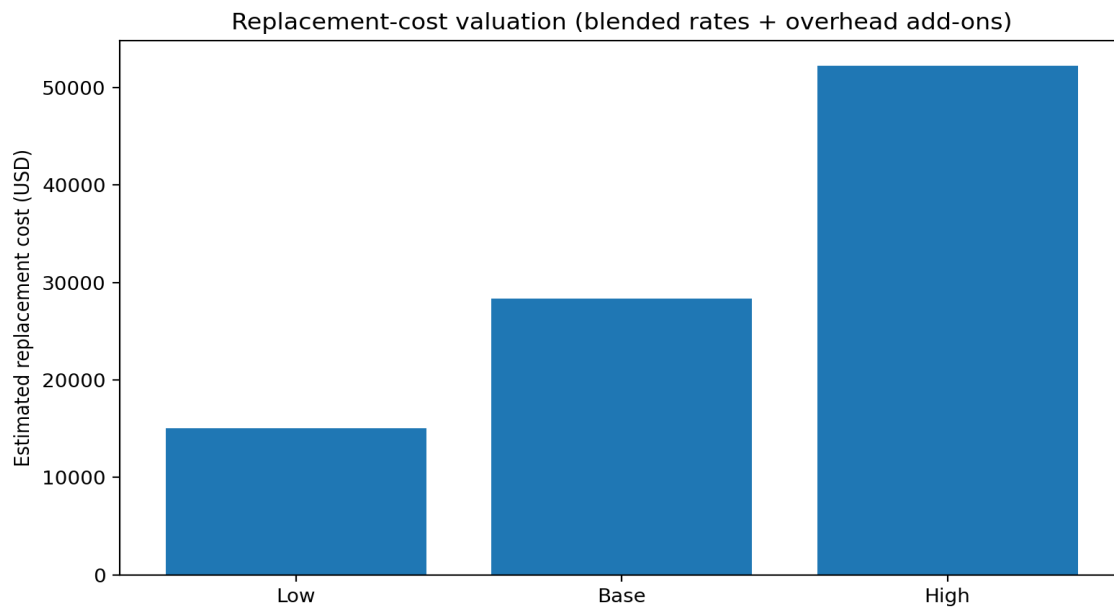
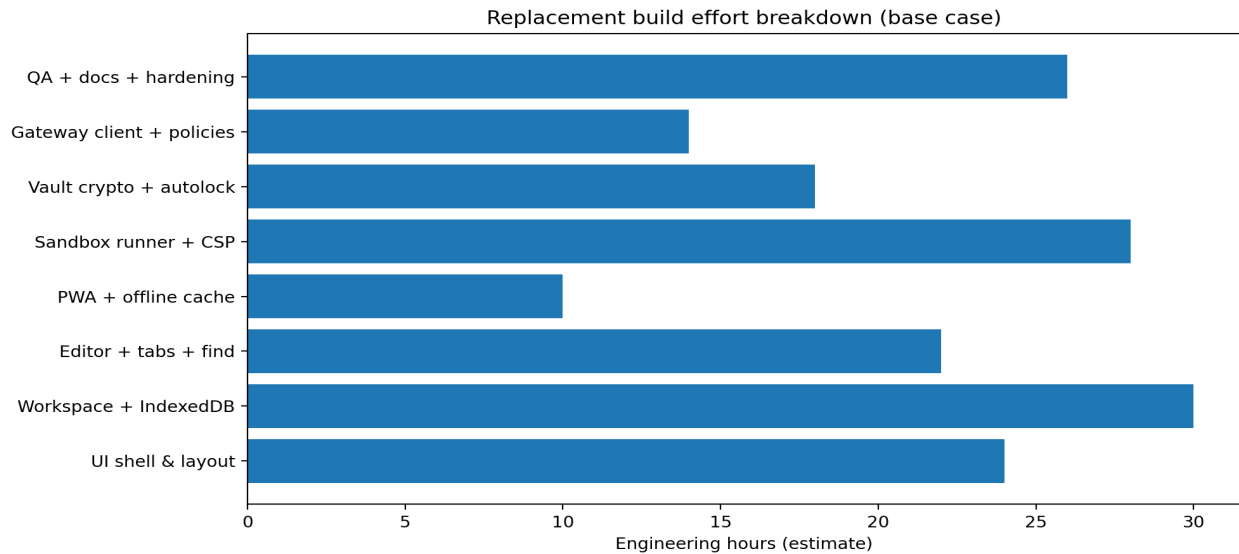
Risk	Likelihood	Impact	Notes / mitigation
Large logo asset increases first-load latency	High	Medium	Optimize assets; consider SVG/WebP; lazy-load.
HTML sanitizer is basic (preview mode)	Medium	Medium	For enterprise, use a hardened sanitizer or disable preview for untrusted HTML.
Client-only: no centralized auth/RBAC	High	High	Add OIDC/SSO, orgs, RBAC for enterprise rollouts.
Client-only: no server-side audit trail	High	High	Add immutable server audit log + export.
No automated tests / CI	Medium	Medium	Add unit tests for vault/IDB; smoke tests for SW update flows.

4) Valuation (three lenses)

Valuation depends on whether you are valuing the shipped hardened PWA as an asset, or an enterprise product after adding IAM, RBAC, server audit, and compliance. The numbers below separate those cases and make assumptions explicit.

4.1 Replacement-cost valuation (rebuild cost)

Base-case effort breakdown and cost scenarios:



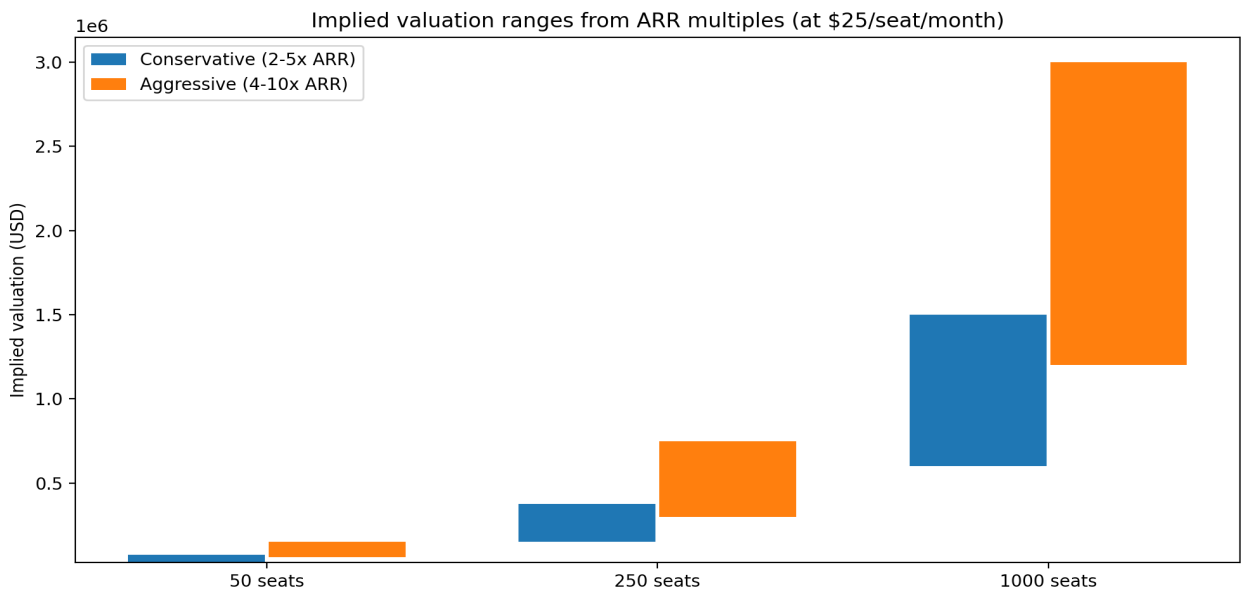
Replacement cost scenarios (table)

Scenario	Hours	Rate (USD/hr)	Replacement cost (USD)
Low	120	\$90.00	\$15,040
Base	172	\$140.00	\$28,320
High	240	\$200.00	\$52,240

As-built replacement-cost range: roughly \$15k-\$52k depending on rate assumptions and QA/hardening intensity. This is a floor, not a market price.

4.2 Revenue-multiple valuation (if sold per seat)

If sold as a developer tool, a common shorthand is an ARR multiple. Multiples vary with growth and retention; conservative private valuations are often lower than public SaaS medians. Example uses \$25/seat/month.



Seats	Price/seat/mo	ARR	2-5x range	4-10x range
50	\$25.00	\$15,000	\$30,000 - \$75,000	\$60,000 - \$150,000
250	\$25.00	\$75,000	\$150,000 - \$375,000	\$300,000 - \$750,000
1000	\$25.00	\$300,000	\$600,000 - \$1,500,000	\$1,200,000 - \$3,000,000

Example: 250 seats at \$25/seat/month is \$75k ARR. At 2-5x ARR that is \$150k-\$375k; at 4-10x ARR that is \$300k-\$750k.

4.3 Strategic / internal asset value

If primarily used internally across your Kaixu toolchain, value is best modeled as avoided engineering time and reduced operational risk (consistent gateway routing, safer local secret handling, standardized offline workspace). Replacement cost is the primary measurable component.

5) Enterprise gap analysis (procurement blockers)

Client-side hardening is necessary but not sufficient for large enterprise adoption. Typical blockers are identity, authorization, centralized logging, and compliance evidence. The table below scopes a practical delta.

Gap item	Effort (hours)	Why it matters
SSO/OIDC login + org users	80-140	Enterprise procurement gate
RBAC + admin console	60-120	Owner/admin vs user; policy enforcement
Server-side audit log + SIEM export	60-100	Central traceability
Central key management + rotation	60-120	Eliminate local-only secrets
Automated tests + CI pipeline	40-80	Regression control
Accessibility (WCAG) pass	16-40	Procurement and legal
Performance: image optimization + lazy load	4-12	First load latency reduction
SOC2 readiness controls + evidence	80-200	Trust and sales enablement

References (for rate and multiple assumptions)

1. US Bureau of Labor Statistics (BLS), Software Developers: median annual wage \$133,080 (May 2024).
<https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm>
2. FullStack Labs, 2025 software development hourly rates (US): \$90-\$160/hour.
<https://www.fullstack.com/labs/resources/blog/software-development-price-guide-hourly-rate-comparison>
3. ContractRates.fyi (crowdsourced), Senior Software Engineer hourly rate (US): about \$115/hour average.
<https://www.contractrates.fyi/Senior-Software-Engineer/hourly-rates>
4. SaaS Capital (Aug 2025), valuation ranges discussed for public SaaS companies (median around 6x-8x ARR; broader 4x-10x).
<https://www.saas-capital.com/blog-posts/saas-valuation-multiples-understanding-the-new-normal/>
5. Public SaaS Companies (Feb 2026), public SaaS revenue multiple snapshot. <https://publicsaascompanies.com/saas-multiples/>